

It's like trying every possible combination on a lock to find the one that opens it.

```
1. def grid_search_optimization(ticker, start_date, end_date, param_grid):
2.     """
3.     Perform grid search optimization for a moving average crossover strategy
4.
5.     Parameters:
6.     ticker (str): Stock symbol
7.     start_date (str): Start date in 'YYYY-MM-DD' format
8.     end_date (str): End date in 'YYYY-MM-DD' format
9.     param_grid (dict): Dictionary with parameter names as keys and lists of values as values
10.
11.     Returns:
12.     pd.DataFrame: Results of all parameter combinations
13.     """
14.     # Prepare data once
15.     data = prepare_backtest_data(ticker, start_date, end_date)
16.
17.     # Create a list to store results
18.     results = []
19.
20.     # Generate all parameter combinations
21.     from itertools import product
22.     param_combinations = list(dict(zip(param_grid.keys(), values))
23.                                for values in product(*param_grid.values()))
24.
25.     print(f"Testing {len(param_combinations)} parameter combinations...")
26.
27.     # Test each parameter combination
28.     for params in param_combinations:
29.         # Extract parameters
30.         short_ma = params['short_ma']
31.         long_ma = params['long_ma']
32.
33.         # Calculate indicators
34.         data['SMA_Short'] = data['Adj Close'].rolling(window=short_ma).mean()
35.         data['SMA_Long'] = data['Adj Close'].rolling(window=long_ma).mean()
36.
```